

# Mathematical Expectation and Generating Function

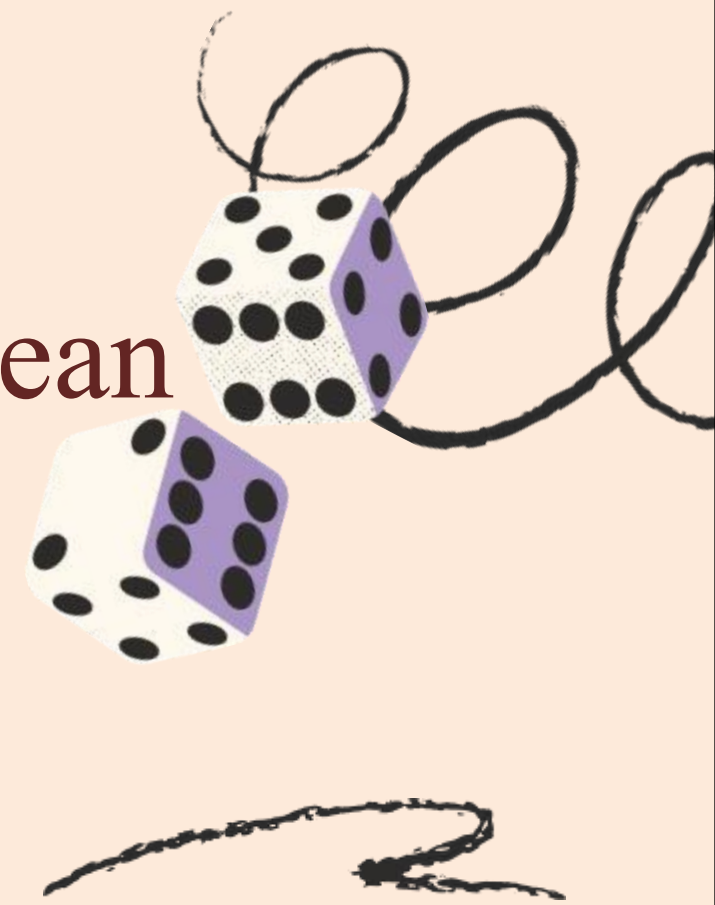


# Expectation of Random Variable

If  $X$  is a discrete random variable assuming values  $x_1, x_2, \dots, x_n$  with corresponding probabilities  $P_1, P_2, \dots, P_n$  then the mathematical expectation of random variable  $X$  is denoted by  $E(X)$  and is denoted by

$$\underline{E(X) = \sum x_i P_i}$$

The expected value of a random variable is the arithmetic mean of that variable,  
i.e.  $E(X) = \mu$

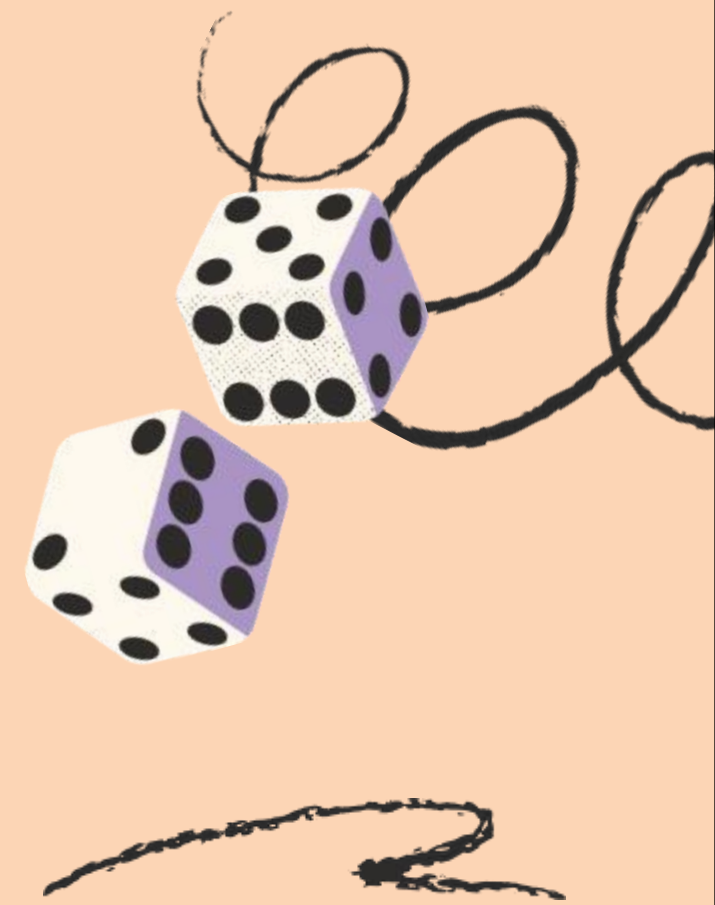


## For Discrete :

The expected value of a discrete random variable,  $X$ , is found by multiplying each  $X$ -value by its probability and then summing over all values of the random variable.

That is, if  $X$  is discrete,

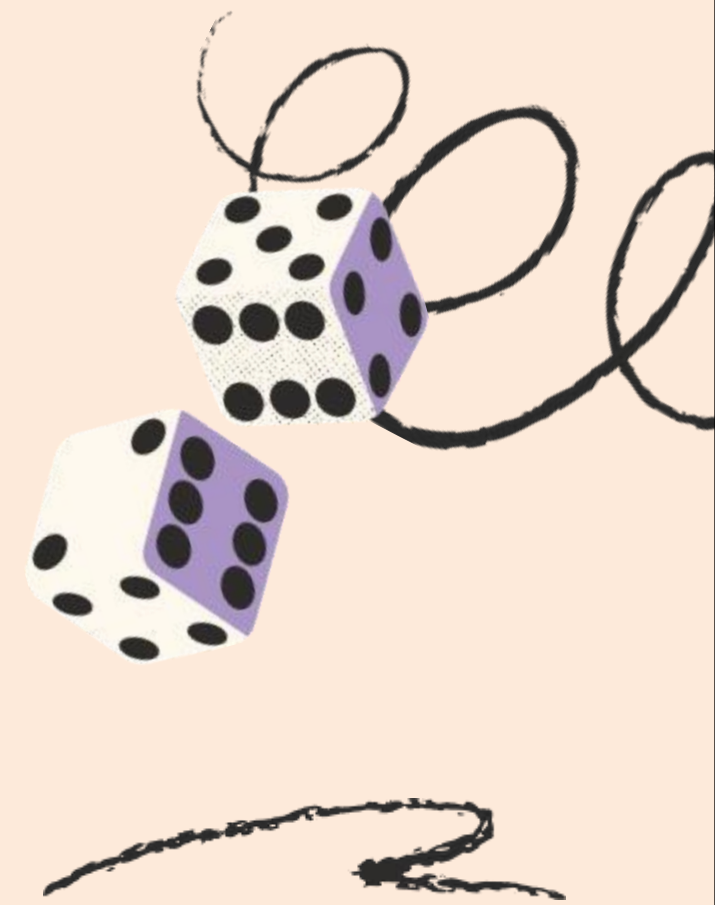
$$\mathbf{E(X) = \sum_{for\ all\ X} xP(x) = \mu_x}$$



## For Continuous :

For a continuous variable  $X$  ranging over all the real numbers, the expectation is defined by

$$\mathbf{E(X)} = \int_{-\infty}^{\infty} \mathbf{x} \mathbf{f(x)} \, d\mathbf{x}$$



# Properties of Expectation

➤  $E(aX) = a E(X)$

where  $a$  is real constant

➤  $E(aX+b) = aE(X) + b$

where  $a$  and  $b$  are real constants

➤  $E(X+Y) = E(X) + E(Y)$

➤  $E(XY) = E(X) \cdot E(Y)$

## Variance Of Random Variable

The variance of a random variable  $X$  is defined as the expected squared deviation of the values of this random variable about their mean

$$V(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$



## For Discrete

The variance of a random variable  $X$  is defined as the expected squared deviation of the values of this random variable about their mean

$$V(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2 = \sum x^2 P(X) - [\sum x P(X)]^2$$

## For Continuous

$$V(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$

*where*  $E(X^2) = \int_{-\infty}^{\infty} x^2 \mathbf{f(x)} \, \mathbf{dx}$

*where*  $f(x)$  is the probability density function



# Properties of Variance

➤  $V(X+a) = V(X)$

➤  $V(aX) = a^2V(X)$

➤  $V(aX+b) = a^2V(X)$

where  $a$  is real constant

where  $a$  and  $b$  are real constants

## Examples:

Q.1 For a random variable  $X$ ,  $E(X) = 10$  &  $V(X) = 5$  ;

Find

- $E(5X-4)$  and  $E(4X+3)$
- $V(3X+5)$ ,  $V(X-2)$  and  $V(4X)$

Q.2

1. If  $E(x) = 3$ ,  $V(x) = 5$  then find  $E(Y)$  and  $V(Y)$  for  $Y = 3x + 4$

2. Given probability function

| $x$    | 0   | 1   | 2   | 3   |
|--------|-----|-----|-----|-----|
| $P(x)$ | 0.1 | 0.3 | 0.5 | 0.1 |

Find  $E(x)$  and  $V(x)$ .

$$1. E(X) = 10 \quad V(X) = 5$$

$$\begin{aligned} E(5X-4) &= E(5X) - E(4) \\ &= 5E(X) - 4 \\ &= 5(10) - 4 \quad \because E(X) = 10 \\ &= 46 \end{aligned}$$

$$\begin{aligned} E(4X+3) &= E(4X) + E(3) \\ &= 4E(X) + 3 \\ &= 4(10) + 3 \\ &= 43 \end{aligned}$$

$$\begin{aligned} V(3X+5) &= V(3X) + V(5) \\ &= 3^2 V(X) + 0 \quad \because V(X) = 5 \\ &= 3^2 (5) \\ &= 9 \times 5 \\ &= 45 \end{aligned}$$

$$\begin{aligned} V(X-2) &= V(X) + V(2) \\ &= 5 + 0 \\ &= 5 \end{aligned}$$

$$\begin{aligned} V(4X) &= 4^2 V(X) = 16(5) \quad \because V(X) = 5 \\ &= 80 \end{aligned}$$

Q.2

1.  $E(X) = 3$

$V(X) = 5$

Find  $E(Y)$  and  $V(Y)$

Given  $Y = 3X + 4$

$$E(Y) = E(3X + 4)$$

$$= E(3X) + E(4)$$

$$= 3E(X) + 4$$

$$= 3(5) + 4$$

$$\therefore E(X) = 5$$

$$E(Y) = 19$$

$$V(Y) = V(3X + 4)$$

$$= V(3X) + V(4)$$

$$= 3^2 V(X) + V(4)$$

$$= 9(5) + 0 \quad \therefore V(X) = 5$$

$$V(Y) = 45$$

|      |     |     |     |     |
|------|-----|-----|-----|-----|
| X    | 0   | 1   | 2   | 3   |
| P(X) | 0.1 | 0.3 | 0.5 | 0.1 |

$$E(X) = \sum_{x \in X} x P(X=x)$$

$$= \sum_{x=0}^3 x P(X)$$

$$= 0 \times 0.1 + 1 \times 0.3 + 2 \times 0.5 + 3 \times 0.1$$

$$E(X) = 1.6$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum_{x \in X} x^2 P(X=x)$$

$$= \sum_{x=0}^3 x^2 P(X)$$

$$= 0^2 \times 0.1 + 1^2 \times 0.3 + 2^2 \times 0.5 + 3^2 \times 0.1$$

$$= 0.3 + 2 + 0.9$$

$$E(X^2) = 3.2$$

$$\therefore V(X) = E(X^2) - [E(X)]^2$$

$$= 3.2 - (1.6)^2 \quad \because E(X) = 1.6$$

$$= 0.64$$

## Examples:

Q.1 Let  $X$  be a continuous random variable with pdf

$$f_x(x) = \begin{cases} x + \frac{1}{2} & ; 0 < x < 1 \\ 0 & \text{o.w} \end{cases}$$

Find Expected Value and Variance

Q.2 Let  $X$  be a continuous random variable with pdf

$$f_x(x) = \begin{cases} \frac{3}{x^4} & ; 1 < x < 4 \\ 0 & \text{o.w} \end{cases}$$

$$Q.1 \quad f(x) = x + \frac{1}{2} \quad 0 < x < 1$$

$$= 0 \quad 0 \leq x$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x \left(x + \frac{1}{2}\right) dx$$

$$= \int_0^1 x^2 + \frac{1}{2}x dx$$

$$= \left[ \frac{x^3}{3} + \frac{1}{2} \cdot \frac{x^2}{2} \right]_0^1$$

$$= \left[ \frac{1}{3} + \frac{1}{2} \left( \frac{1}{2} \right) \right] - 0$$

$$= \frac{1}{3} + \frac{1}{4}$$

$$E(X) = \frac{7}{12}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx$$

$$= \int_0^1 x^3 + \frac{1}{2}x^2 dx$$

$$= \left[ \frac{x^4}{4} + \frac{1}{2} \left( \frac{x^3}{3} \right) \right]_0^1$$

$$= \left[ \frac{1}{4} + \frac{1}{2} \left( \frac{1}{3} \right) \right] - 0$$

$$= \frac{1}{4} + \frac{1}{6}$$

$$E(X^2) = \frac{10}{24}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \frac{10}{24} - \left( \frac{7}{12} \right)^2 \quad \because E(X) = \frac{7}{12}$$

$$= \frac{11}{144}$$

$$= 0.076$$

$$Q.2 \ f(x) = \frac{3}{x^4} \quad 1 < x < 4$$

$$E(X) = \int_1^4 x f(x) dx$$

$$= \int_1^4 x \frac{3}{x^4} dx$$

$$= \int_1^4 \frac{3}{x^3} dx = \int_1^4 3x^{-3} dx$$

$$= \frac{3x^{-2}}{-2} \Big|_1^4$$

$$= -\frac{3}{2x^2} \Big|_1^4$$

$$= -\frac{3}{2(4)^2} - \left( -\frac{3}{2(1)^2} \right)$$

$$= -\frac{3}{32} + \frac{3}{2}$$

$$= \frac{45}{32}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_1^4 x^2 f(x) dx$$

$$= \int_1^4 x^2 \frac{3}{x^4} dx$$

$$= \int_1^4 \frac{3}{x^2} dx = \int_1^4 3x^{-2} dx$$

$$= \frac{3x^{-1}}{-1} \Big|_1^4$$

$$= -\frac{3}{x} \Big|_1^4$$

$$= -\frac{3}{4} - \left( -\frac{3}{1} \right)$$

$$= -\frac{3}{4} + 3$$

$$= \frac{9}{4}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \frac{9}{4} - \left( \frac{45}{32} \right)^2$$

$$= 0.2724$$

# Conditional Mean

- $E(X|Y=y) = \sum_x x P(x|y)$
- $E(Y|X=x) = \sum_y y P(y|x)$

**For X and Y with joint pmf given below  $E(Y \mid X=2)$  and  $V(Y \mid X=2)$**

| <div><div></div><div>Y</div><div>X</div></div> | 1   | 2   | 3   | Total |
|------------------------------------------------|-----|-----|-----|-------|
| 1                                              | 0.2 | 0.3 | 0.1 | 0.6   |
| 2                                              | 0.2 | 0.1 | 0.1 | 0.4   |
| Total                                          | 0.4 | 0.4 | 0.2 | 1     |

# Solution

| Q | $X \backslash Y$ | 1   | 2   | 3   | Total $P(X=x)$ |
|---|------------------|-----|-----|-----|----------------|
|   | 1                | 0.2 | 0.3 | 0.1 | 0.6            |
|   | 2                | 0.2 | 0.1 | 0.1 | 0.4            |
|   | $P(Y=y)$ Tot     | 0.4 | 0.4 | 0.2 | 1              |

$$E(Y|X=2) = \sum_y y P(Y|X=2)$$

$$\therefore P(Y|X) = \frac{P(Y \cap X)}{P(X=X)}$$

| Y          | 1       | 2       | 3       |
|------------|---------|---------|---------|
| $P(Y X=2)$ | 0.2/0.4 | 0.1/0.4 | 0.1/0.4 |

| Y          | 1   | 2   | 3   |
|------------|-----|-----|-----|
| $P(Y X=2)$ | 2/4 | 1/4 | 1/4 |

$$\therefore E(Y|X=2) = \sum_{y=1}^3 y P(Y|X=2)$$

$$= 1 \times \frac{2}{4} + 2 \times \frac{1}{4} + 3 \times \frac{1}{4}$$

$$= \frac{2}{4} + \frac{2}{4} + \frac{3}{4}$$

$$E(Y|X=2) = \frac{7}{4}$$

$$V(Y|X=2) = E(Y^2|X=2) - [E(Y|X=2)]^2$$

$$E(Y^2|X=2) = \sum_{y \in Y} y^2 P(Y|X=2)$$

$$= 1^2 \times \frac{2}{4} + 2^2 \times \frac{1}{4} + 3^2 \times \frac{1}{4}$$

$$= \frac{2}{4} + \frac{4}{4} + \frac{9}{4}$$

$$= \frac{15}{4}$$

$$V(Y|X=2) = E(Y^2|X=2) - [E(Y|X=2)]^2$$

$$= \frac{15}{4} - \left(\frac{7}{4}\right)^2$$

$$= \frac{11}{16}$$

# Moment Generating Function

The moment generating function of Random Variable  $X$  is given by

## For Discrete

$$M_X(t) = E(e^{tX}) = \sum_{-\infty}^{\infty} e^{tx} \cdot P_X(x) \quad (\text{Discrete})$$

## For Continuous

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) \cdot dx \quad (\text{Continuous})$$

## Properties of MGF:

- If  $M_X(t)$  is the moment generating function of random variable  $X$  then the MGF of variable  $cX$  defined as  $\mathbf{M}_{cX}(\mathbf{t}) = \mathbf{M}_X(\mathbf{ct})$
- If  $X$  and  $Y$  are independent random variables with MGFs  $M_X(t)$  and  $M_Y(t)$ , respectively, then the MGF of their sum  $(X + Y)$  is the product of their MGFs:  $M_{X+Y}(t) = M_X(t) * M_Y(t)$ .
- If  $U = \frac{X-a}{h}$   $\mathbf{M}_u(\mathbf{t}) = \mathbf{E}(\mathbf{e^{tu}}) = \mathbf{e^{\frac{-ta}{h}}} M_X(\frac{t}{h})$
- If  $M_X(t) = M_Y(t)$  then  $X=Y$

**NOTE:**

- It can be used to easily derive moments; its derivatives at zero are equal to the moments of the random variable
- The probability distribution is uniquely determined by its MGF.
- The  $r$ th derivative of the mgf evaluated at  $t=0$  gives the value of the  $r$ th moment.

Find MGF of a random variable  $X$  whose Probability function is

$$\text{a. } P(X) = \frac{1}{3^x} \quad ; x = 1, 2, 3 \\ 0 \quad \text{o.w}$$

$$\text{b. } P(X) = 1/4 \quad x=0,1,2 \\ = 0 \quad \text{o.w}$$

$$\text{c. } f(X) = \frac{1}{(b-a)} \quad a < x < b$$

$$\text{d. } f(X) = \lambda e^{-\lambda x} \quad x > 0$$

# SOLUTION

$$\begin{aligned} \text{a) } P(X) &= \frac{1}{3^x} \quad x=1,2,3 \\ &= 0 \quad \text{o.w.} \end{aligned}$$

$$\begin{aligned} M_X(t) &= E(e^{tx}) \\ &= \sum_{x=1}^3 e^{tx} P(x) \end{aligned}$$

$$= \sum_{x=1}^3 e^{tx} \frac{1}{3^x}$$

$$= \sum_{x=1}^3 \left(\frac{e^t}{3}\right)^x$$

$$M_X(t) = \left(\frac{e^t}{3}\right)^1 + \left(\frac{e^t}{3}\right)^2 + \left(\frac{e^t}{3}\right)^3$$

$$\begin{aligned} \text{b) } P(X) &= \frac{1}{4} \quad x=0,1,2 \\ &= 0 \quad \text{o.w.} \end{aligned}$$

$$M_X(t) = E(e^{tx})$$

$$= \sum_{x=0}^2 e^{tx} P(x)$$

$$= \sum_{x=0}^2 e^{tx} \frac{1}{4}$$

$$= \frac{1}{4} [e^{t(0)} + e^{t(1)} + e^{t(2)}]$$

$$M_X(t) = \frac{1}{4} [1 + e^t + e^{2t}]$$

# SOLUTION

$$c) f(x) = \frac{1}{b-a} \quad a < x < b$$

$$M_X(t) = E(e^{tx})$$

$$= \int_a^b e^{tx} f(x) dx$$

$$= \int_a^b e^{tx} \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^b e^{tx} dx$$

$$= \frac{1}{b-a} \left. \frac{e^{tx}}{t} \right|_a^b$$

$$= \frac{1}{b-a} \left[ \frac{e^{tb}}{t} - \frac{e^{ta}}{t} \right]$$

$$= \frac{1}{(b-a)t} e^{t(b-a)}$$

$$M_X(t) = \frac{1}{(b-a)t} e^{t(b-a)}$$

$$d) f(x) = \lambda e^{-\lambda x} \quad x > 0$$

$$M_X(t) = E(e^{tx})$$

$$= \int_0^{\infty} e^{tx} f(x) dx$$

$$= \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} e^{(t-\lambda)x} dx$$

$$= \lambda \left. \frac{e^{(t-\lambda)x}}{(t-\lambda)} \right|_0^{\infty}$$

$$= \lambda \left[ \frac{e^{\infty}}{t-\lambda} - \frac{e^0}{t-\lambda} \right]$$

$$= \lambda \left[ 0 - \frac{1}{t-\lambda} \right]$$

$$= -\frac{\lambda}{t-\lambda}$$

$$M_X(t) = \frac{\lambda}{\lambda - t}$$

## Cumulant Generating Function

The cumulant generating function of a random variable is the natural logarithm of its moment generating function.

Suppose that a random variable  $X$  possesses a moment generating function

Then, the function

$$\mathbf{K_X(t) = \ln[M_X(t)]}$$

is the cumulant generating function of  $X$ .

## Cumulants :

**The derivatives of the cgf at zero are called cumulants.**

The  $n^{\text{th}}$  cumulant is

$$\kappa_X(n) = \left. \frac{d^n K_X(t)}{dt^n} \right|_{t=0}$$

- The first cumulant  $K_X(1)$  =  $E(X)$  = Mean
- The second cumulant  $K_X(2)$  =  $E(X-E(X))^2 = V(X)$  = 2nd central Moment
- The third cumulant  $K_X(3)$  =  $E(X-E(X))^3$  = 3rd Central Moment

## Properties of CGF

1. If  $X_1, X_2, \dots, X_n$  be the independent random variable with CGF  $K_{X_1}(t), K_{X_2}(t), \dots, K_{X_n}(t)$  then the CGF of  $X_1 + X_2 + \dots + X_n$  is given by

$$K_{X_1 + X_2 + \dots + X_n}(t) = K_{X_1} + K_{X_2} + \dots + K_{X_n}$$

2. If  $U = \frac{X-a}{h}$  then  $K_u(t) = \frac{-ta}{h} + \log M_x\left(\frac{t}{h}\right)$

$$M_u(t) = E(e^{tu}) = e^{\frac{-ta}{h}} M_x\left(\frac{t}{h}\right)$$

# Properties of CGF

1. If  $X_1, X_2, \dots, X_n$  be the independent random variable with CGF  $K_{X_1}(t), K_{X_2}(t), \dots, K_{X_n}(t)$  then the CGF of  $X_1 + X_2 + \dots + X_n$  is given by  $K_{X_1+X_2+\dots+X_n}(t) = K_{X_1} + K_{X_2} + \dots + K_{X_n}$

$$1. \quad K_{X_1+X_2+\dots+X_n}(t) = K_{X_1} + K_{X_2} + \dots + K_{X_n}$$

Let  $X_1, X_2, \dots, X_n$  be independent random variable

$$\text{Let } K_X(t) = \ln [M_X(t)]$$

$$\therefore K_{X_1+X_2+\dots+X_n}(t) = \ln [M_{X_1+X_2+\dots+X_n}(t)] \quad \text{--- (i)}$$

Since  $M_{X+Y} = M_X(t) \times M_Y(t)$  ;  $X$  and  $Y$  are independent

$\therefore$  from (i)

$$\begin{aligned} K_{X_1+X_2+\dots+X_n}(t) &= \ln [M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t)] \\ &= \ln M_{X_1}(t) + \ln M_{X_2}(t) + \dots + \ln M_{X_n}(t) \\ &= K_{X_1}(t) + K_{X_2}(t) + \dots + K_{X_n}(t) \end{aligned}$$

# Properties of CGF

2. If  $U = \frac{X-a}{h}$  then  $K_U(t) = \frac{-ta}{h} + \log M_X\left(\frac{t}{h}\right)$

$$M_U(t) = E(e^{tu}) = e^{\frac{-ta}{h}} M_X\left(\frac{t}{h}\right)$$

2.  $U = \frac{X-a}{h}$  then  $K_U(t) = -\frac{ta}{h} + \log M_X\left(\frac{t}{h}\right)$

$$\therefore K_U(t) = \ln[M_U(t)]$$

$$\therefore M_U(t) = M_{\frac{X-a}{h}}(t)$$

$$= e^{\frac{-ta}{h}} M_X\left(\frac{t}{h}\right)$$

$$\therefore K_U(t) = \ln(M_U(t))$$

$$= \ln\left(M_{\frac{X-a}{h}}(t)\right)$$

$$= \ln\left(e^{\frac{-ta}{h}} M_X\left(\frac{t}{h}\right)\right)$$

$$= \ln e^{\frac{-ta}{h}} + \ln M_X\left(\frac{t}{h}\right)$$

$$K_U(t) = -\frac{ta}{h} + \ln M_X\left(\frac{t}{h}\right)$$

# Probability Generating Function

Let  $X$  be a discrete random variable with values  $X_1, X_2, \dots, X_n$  then the PGF is given by

$$G_X(s) = \mathbb{E}(s^X) = \sum_{x=0}^{\infty} s^x \mathbb{P}(X = x).$$

## Properties of the PGF:

### 1. $G_X(0) = \mathbb{P}(X = 0)$ :

$$\begin{aligned} G_X(0) &= 0^0 \times \mathbb{P}(X = 0) + 0^1 \times \mathbb{P}(X = 1) + 0^2 \times \mathbb{P}(X = 2) + \dots \\ \therefore G_X(0) &= \mathbb{P}(X = 0). \end{aligned}$$

### 2. $G_X(1) = 1$ :

$$G_X(1) = \sum_{x=0}^{\infty} 1^x \mathbb{P}(X = x) = \sum_{x=0}^{\infty} \mathbb{P}(X = x) = 1.$$